

Degree of Relationship & Cardinality

Topic: 1:1 Relationship

Exam Importance: High ★★★★★

1. What is a Relationship?

In an **Entity-Relationship (ER) Model**, a relationship represents an association or connection between two or more entities.

STUDENT ----- < STUDIES > ----- COURSE

- **STUDENT:** Entity Set
- **COURSE:** Entity Set
- **STUDIES:** Relationship set linking students to their enrolled courses

2. Types of Relationships / Cardinality Ratio

Cardinality defines the maximum number of entity instances that can be associated with another entity instance in a relationship. The major types are:

- **One-to-One (1:1)**
- **One-to-Many (1:N)**
- **Many-to-One (N:1)**
- **Many-to-Many (M:N)**

! Key Exam Notation Rule: For Many-to-Many relationships, always write **M:N** (or **1:N / N:1** for semi-many), never **M:M**. This indicates that the exact number of entities on each side does not have to be equal.

3. One-to-One Relationship (1:1) ★★★★★

A **One-to-One (1:1) relationship** dictates that an entity in set A is associated with at most one entity in set B, and an entity in set B is associated with at most one entity in set A.

EMPLOYEE ----- [1 : 1] ----- DEPARTMENT

Example Constraint:

- One employee can manage/work in at most **one** department.
- One department can have at most **one** employee as a manager/member.

Mapping: E1 → D1, E2 → D3, E3 → D2. An employee cannot simultaneously be assigned to two departments under this strict constraint.

4. Entity Tables Conversion

Suppose we represent the two entities as individual relational tables:

EMPLOYEE Table

EID PK	Name	Age
E1	A	20
E2	B	25
E3	C	28
E4	A	24

DEPARTMENT Table

DID PK	Department_Name	Location
D1	IT	Bangalore
D2	Production	Delhi
D3	HR	Delhi

5. Relationship Table (Initial State)

Initially, the relationship WORKS_IN can be created as an independent schema carrying foreign keys from both participating entities, alongside any descriptive attributes.

WORKS_IN Table

EID FK	DID FK	Joining_Date (Descriptive Attribute)
E1	D1	2026-01-10
E2	D3	2026-01-15
E3	D2	2026-02-01
E4	NULL	2026-02-10

6. Uniqueness & Key Constraints

Why can't EID or DID be repeated in a strict 1:1 relationship table?

- If **E1** → **D1** and **E1** → **D2** exist, then employee E1 works in two departments, violating 1:1 cardinality.
- If **E1** → **D1** and **E2** → **D1** exist, then department D1 has two employees, violating 1:1 cardinality.

Conclusion: Neither EID nor DID can contain duplicate values in the relationship table.

7. Primary Key Selection for Relationship Table ★★★★★

Exam Key Point: In a 1:1 relationship, because both participating key attributes (EID and DID) are strictly unique, **either key can act as a Candidate Key** and be chosen as the Primary Key for the relationship table.

- **Option 1:** Select EID as Primary Key, DID as Foreign Key.
- **Option 2:** Select DID as Primary Key, EID as Foreign Key.

8. Schema Reduction: Merging Tables ★★★★★

Initially, an ER diagram yields **3 tables** (Entity A, Entity B, and Relationship R). However, because the cardinality is 1:1, we can optimize space and joins by merging the relationship with either entity table.

Approach A: Merge WORKS_IN into EMPLOYEE

If we choose EID as the Primary Key of the relationship, we combine EMPLOYEE + WORKS_IN:

Merged EMPLOYEE Table

EID PK	Name	Age	DID FK
E1	A	20	D1
E2	B	25	D3
E3	C	28	D2
E4	A	24	NULL

Why can DID contain NULL?

If employee E4 is not yet assigned to any department, DID can be NULL. This is valid because DID is acting as a foreign key here. The primary key (EID) strictly remains non-null.

Approach B: Merge WORKS_IN into DEPARTMENT

Alternatively, if we choose DID as the Primary Key, we combine DEPARTMENT + WORKS_IN. The final schemas remain reduced to **2 tables** regardless of the choice made.

9. Summary Diagram & Conversion Workflow

[EMPLOYEE] --- (1:1) --- [DEPARTMENT]

↓

Initial: EMPLOYEE (1) + DEPARTMENT (2) + WORKS_IN (3)

↓ (Apply 1:1 Merge Rule)

Final: 2 Tables (EMPLOYEE + DEPARTMENT)

🎯 High-Yield Exam Questions

Q1. What are the major types of cardinality in DBMS?

Answer: 1:1 (One-to-One), 1:N (One-to-Many), N:1 (Many-to-One), and M:N (Many-to-Many).

Q2. Can a 1:1 relationship table contain duplicate values in Foreign Key attributes?

Answer: No. Duplicate values on either side violate the one-to-one mapping constraint.

Q3. What can be selected as the Primary Key of a 1:1 relationship table?

Answer: Either participating entity's primary key (e.g., EID or DID) can serve as the Primary Key since both are Candidate Keys.

Q4. What is the minimum number of tables required to represent a 1:1 ER relationship in a relational schema?

Answer: 2 tables (by merging the relationship table into either of the entity tables).

 Golden Rules for Exams ★★★★★

1. **1:1 Cardinality:** Neither side contains duplicate entries in the relationship mapping.
2. **Keys Included:** The relationship table initially holds primary keys of both participating entities.
3. **Candidate Keys:** Both foreign keys act as valid candidate keys in a strict 1:1 relation.
4. **Primary Key Choice:** Either entity's key can be designated as the Primary Key.
5. **Table Reduction:** Merging the relationship with either entity reduces total tables from **3 to 2**.

 **One-Line Memory Trick: 1:1 → Both Unique → Either can be PK → Merge with either side → 2 Tables!**